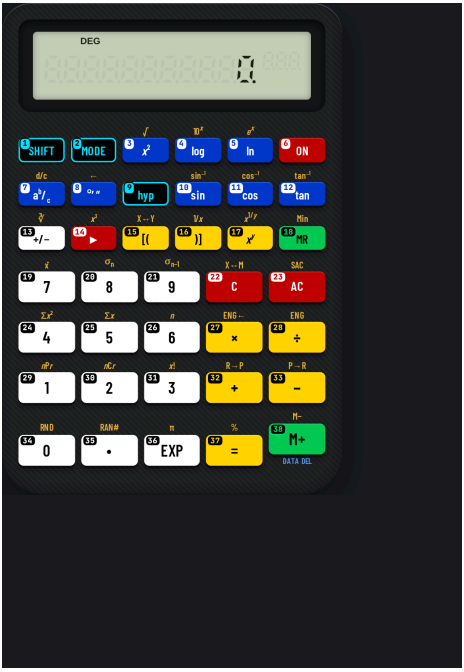
U-45 CALCULATORS

User Manual

Calculator #23 Classic One-Line Scientific (textured black)

Scientific – classic one-line



Complete kit: functions · numbered key map · worked examples · tricks & hacks · shortcuts · common mistakes · practice drills

About this manual

This manual belongs to the **U-45 CALCULATORS** app and describes calculator **#23 — Classic One-Line Scientific (textured black)** exactly as it behaves in the app. Every worked example and every drill answer in this book was run on the app's own keys by an automated checker before printing; only results that matched were kept.

How key presses are written

Keys are written as a chip showing the **key number** and its face, for example 

. Numbers follow the numbering printed on the keys (switch it on with "# On"). A shifted function is written as two chips: the shift key, then the key that carries the function.

Entry style of this model

AOS algebraic (operator precedence, = completes).

Chapters

1. Getting started
2. Features
3. Numbered key map
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5. Worked examples
6. Tricks & hacks
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8. Common mistakes
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1 · Getting started

Open U-45 CALCULATORS, choose **#23** from the home screen (or the jump list at the top), and the calculator appears with its numbered, colour-coded keys. Click or tap keys, or type on a keyboard.

Numbers on every key

Every key carries a small number in the same reading order as this manual (left → right, top → bottom; the navigation pad counts ▲ ▼ ◀ ▶ then its centre). Use "# On / # Off" in the bar above the calculator to show or hide them — the setting is remembered.

High-contrast colours

Keys are coloured by job: white = digits, yellow = operators/execute, blue = functions, green = memory/variables, red = clear/delete/power, black with cyan = shift/modes/menus/navigation. Press "Classic" to return to the original model colours and "Hi-Contrast" to come back.

Three typefaces

Display text uses the U45 Display face, key legends use U45 Keys (a condensed face that stays readable on small keys) and key numbers / helper text use U45 Mono. All three are bundled inside the app, so they work offline.

Copy the display

The Copy button copies exactly what the display shows (indicators that are switched off are skipped) so you can paste a result into a message or spreadsheet.

Fit or 1:1

Fit scales the calculator to your screen; 1:1 shows it at its design size. On a phone, Fit is the comfortable choice.

Reset

Reset rebuilds this calculator from scratch: display, memory, stack and modes return to their start-up state. Use it before each worked example in this manual.

Keyboard

On a computer you can type: digits, + − * / ^ ( ), Enter or = to execute, Backspace to delete, Esc to clear, arrows to move. Letters reach common functions (see the Keyboard shortcuts chapter).

Tooltips

Hover over (or long-press) a key to see "#number name".

Tools rail (v1.4)

A cyan icon rail on the right of every calculator opens 15 panels: Currency (166 currencies, rates auto-updated from open.er-api.com or Frankfurter when online and cached with a date; the bundled dated snapshot is used offline), Units (exact SI factors), Health (published formulas, not medical advice), Simple, Complex, Loans, Construction, Mechanical and Engineering (30+ numbered tools each), Calendar generator, Customize (26 options, saved per calculator), Settings (31 options), History, Hide and View. On narrow phones the rail shows icons only.

History (v1.4)

Every calculation is logged per calculator with a date and time. Open History on the rail to search, re-use (types the input back into the calculator), copy, delete one, select several and delete, or delete all (asks for confirmation). History is stored encrypted on this device (AES-GCM); exporting it needs the passcode.

Camera Solve (v1.4)

The green CS button between Restore and Social reads a problem from the camera, a photo, a PDF or a text file. Recognition (Tesseract OCR, English) runs offline on the device; an online language download is only used if you switch it on in Settings. Check and edit the recognised text, then Solve shows the steps offline. OCR is not available when the app is opened as a plain file:// page in a browser.

Layouts and view modes (v1.4)

The layout selector between Fit and # offers Auto plus 12 layouts: 3D classic, 3D high-contrast, Classic, Pocket, Compact, Expanded, Desktop wide, Split, One-hand left, One-hand right, Large print and Minimal. Auto picks one for your screen and your choice is remembered per calculator. The View panel hides the top bars, the tool bar, the rail or everything (full screen); the round yellow ⤢ button at bottom-left (or Esc) always restores the normal view.

Hide items (v1.4)

The Hide panel lists every button, rail icon and section of this calculator as tick boxes. Hiding is instant; unhiding asks for the app passcode.

Image protection (v1.4)

Right-click, drag and long-press saving of images is blocked inside the app. Saving or sharing an image export (PNG, JPG, QR, calendar), printing a calendar, and downloading or sharing the app itself ask for the passcode, and exported images carry a U-45 watermark. Honest note: this cannot stop a screenshot or screen recording made by the operating system or another device.

Works offline

U-45 CALCULATORS is an offline app: after the first load everything — including fonts — comes from the device.

Colour groups on this calculator

Colour	Group	Keys on this model
num	Digits & number entry	#13, #19, #20, #21, #24, #25, #26, #29, #30, #31, #34, #35, #36
op	Operators & execute	#15, #16, #17, #27, #28, #32, #33, #37
fn	Functions	#3, #4, #5, #7, #8, #10, #11, #12
mem	Memory, variables & registers	#18, #38
clr	Clear, delete & power	#6, #14, #22, #23
mod	Shift, modes, menus & navigation	#1, #2, #9

2 · Features

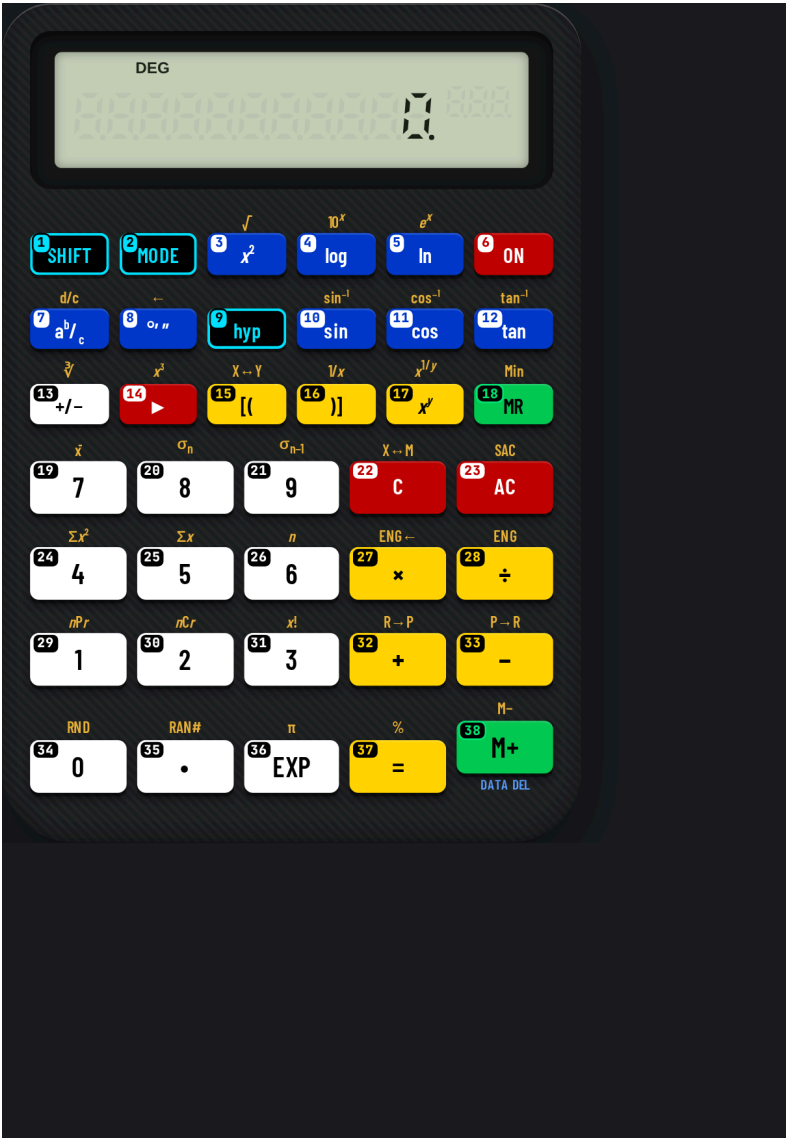
62 features: 43 live on the keys, 9 app/UI features, 10 planned (documented but not yet working — they are clearly marked and never used in examples).

1. LIVE **Classic one-line immediate-execution algebraic logic**
2. LIVE **MODE 0 COMP, MODE 2 SD statistics**
3. LIVE **MODE 4/5/6 = DEG / RAD / GRA, 7/8/9 = FIX / SCI / NORM**
4. LIVE **SD: M+ = DATA, SHIFT M+ = delete; $\bar{x}$, σn, $\sigma n{-}1$, Σx^2, Σx, n**
5. LIVE **a b/c fractions and d/c conversion**
6. LIVE **Constant calculations (K) on repeated operators**
7. LIVE **x ⇄ M exchange**
8. LIVE **S⇔D / F⇔D toggle between exact (fraction, √, π) and decimal answers**
9. LIVE **Square root $\sqrt{}$**
10. LIVE **Cube root $\sqrt[3]{}$**
11. LIVE **Any-index root $\sqrt[x]{y}$**
12. LIVE **Square x^2**
13. LIVE **Cube x^3**
14. LIVE **Reciprocal x^{-1} (1/x)**
15. LIVE **Power x^y / ^ with right-to-left chaining (2^3^2 = 512)**
16. LIVE **Common logarithm $\log_{10}$**
17. LIVE **Natural logarithm ln**
18. LIVE **Antilog 10^x**
19. LIVE **Exponential e^x**
20. LIVE **Trigonometry sin / cos / tan honouring the current angle unit**

21. LIVE Inverse trigonometry $\sin^{-1}$ / $\cos^{-1}$ / $\tan^{-1}$ with domain checking (Math ERROR outside −1…1)
22. LIVE Hyperbolic sinh / cosh / tanh and their inverses via hyp
23. LIVE Sexagesimal ° ′ ″ entry and decimal ↔ DMS conversion
24. LIVE Rectangular → polar conversion Pol(x,y) → r, θ
25. LIVE Polar → rectangular conversion Rec(r,θ) → x, y
26. LIVE Factorial x! (exact up to 69!)
27. LIVE Permutations nPr
28. LIVE Combinations nCr
29. LIVE Random number Ran# (0 ≤ r < 1)
30. LIVE Rnd — round the value to the displayed precision
31. LIVE Percent key (x %, add-on and discount chains)
32. LIVE π constant to full internal precision
33. LIVE Scientific-notation entry ×10^x / EXP / EEX
34. LIVE Sign change (−) / +/− / CHS
35. LIVE Engineering notation ENG (exponent multiple of 3)
36. LIVE ENG← shift the exponent the other way
37. LIVE Store a value into memory / a variable (STO, sto►, →, Min)
38. LIVE Recall memory / variables (RCL, MR, recall)
39. LIVE Independent memory M+
40. LIVE Independent memory M−
41. LIVE MODE / SETUP: Fix / Sci / Norm display, angle unit, fraction format
42. LIVE Backspace / DEL editing of the current entry
43. LIVE Exchange x ⇄ y / x ⇄ M

44. APP/UI Every key numbered 1…N in reading order — matches the key map in the user manual
45. APP/UI High-contrast 6-colour key groups (digits, operators, functions, memory, clear, modes) with a “Classic” toggle
46. APP/UI Three bundled offline fonts: Oxanium (display/titles), Barlow Condensed (key faces), JetBrains Mono (key numbers & helpers)
47. APP/UI Show / hide key numbers with one button (remembered between sessions)
48. APP/UI Copy the current display text to the clipboard
49. APP/UI Physical-keyboard shortcuts: digits, + – * / ^ ( ) . Enter Backspace Esc arrows, s c t l n r ! % e p q i
50. APP/UI Fit / 1:1 zoom that never pushes keys off a phone screen
51. APP/UI Reference photo viewer and tool-tip name on every key
52. APP/UI Reset = fresh power-on state; works fully offline (PWA cache / Android APK)
53. PLANNED Linear regression mode
54. PLANNED Base-N mode
55. PLANNED Replay
56. PLANNED Equation solver
57. PLANNED Two-line display
58. PLANNED Answer history export
59. PLANNED Speech read-out of results
60. PLANNED Colour-blind key theme
61. PLANNED Exam lock
62. PLANNED Custom shortcut keys

3 · Numbered key map



Calculator #23 with key numbers shown (Hi-Contrast colours).

Key table

“Shift” = 2nd / f / SHIFT layer; “Alpha” = ALPHA layer; “3rd” = 3rd / g layer. A dot means the layer is empty on that key.

#	Face	Shift	Alpha	3rd	Group
1	SHIFT	·	·	·	mod
2	MODE	·	·	·	mod
3	x^2	$\sqrt{}$	·	·	fn
4	log	10x	·	·	fn
5	ln	ex	·	·	fn
6	ON	·	·	·	clr
7	ab/c	d/c	·	·	fn
8	°′ ″	←	·	·	fn
9	hyp	·	·	·	mod
10	sin	$\sin^{-1}$	·	·	fn
11	cos	$\cos^{-1}$	·	·	fn
12	tan	$\tan^{-1}$	·	·	fn
13	+/−	$\sqrt[3]{}$	·	·	num
14	▶	x^3	·	·	clr
15	[(	X ↔ Y	·	·	op
16	)]	1/x	·	·	op
17	xy	x1/y	·	·	op
18	MR	Min	·	·	mem
19	7	$\bar{x}$	·	·	num
20	8	σn	·	·	num
21	9	σn−1	·	·	num
22	C	X ↔ M	·	·	clr
23	AC	SAC	·	·	clr

#	Face	Shift	Alpha	3rd	Group
24	4	Σx^2	·	·	num
25	5	Σx	·	·	num
26	6	n	·	·	num
27	×	ENG ←	·	·	op
28	÷	ENG	·	·	op
29	1	nPr	·	·	num
30	2	nCr	·	·	num
31	3	x!	·	·	num
32	+	R → P	·	·	op
33	−	P → R	·	·	op
34	0	RND	·	·	num
35	•	RAN#	·	·	num
36	EXP	π	·	·	num
37	=	%	·	·	op
38	M+	M−	·	·	mem

4 · Key-by-key reference

Every key on calculator #23, in number order, with every function it carries in this app.

#1 SHIFT `Shift, modes, menus & navigation`

Primary: SHIFT / 2nd — the next key uses its upper (shift-coloured) function

#2 MODE `Shift, modes, menus & navigation`

Primary: MODE — choose calculation mode

#3 x^2 `Functions`

Primary: Square x^2

shifted (2nd / f / SHIFT layer) (press #1 first): $\sqrt{}$ — Square root $\sqrt{}$

Keyboard: q

#4 log `Functions`

Primary: Common logarithm $\log_{10}$

shifted (2nd / f / SHIFT layer) (press #1 first): 10x — 10^x — antilog

Keyboard: l

#5 ln `Functions`

Primary: Natural logarithm ln

shifted (2nd / f / SHIFT layer) (press #1 first): ex — e^x — exponential

Keyboard: n

#6 ON `Clear, delete & power`

Primary: ON — power on

#7 ab/c `Functions`

Primary: a b/c — enter fraction / mixed-number parts

shifted (2nd / f / SHIFT layer) (press #1 first): d/c — d/c — switch between mixed number and improper fraction

#8 °′ ″ `Functions`

Primary: Degrees-minutes-seconds entry/display °′ ″

shifted (2nd / f / SHIFT layer) (press #1 first): ← — Convert D°M′ S″ to decimal degrees

#9 hyp `Shift, modes, menus & navigation`

Primary: Hyperbolic prefix — the next sin/cos/tan becomes sinh/cosh/tanh

#10 sin `Functions`

Primary: Sine (uses the current angle unit)

shifted (2nd / f / SHIFT layer) (press #1 first): $\sin^{-1}$ — Inverse sine $\sin^{-1}$

Keyboard: S

#11 cos Functions

Primary: Cosine (current angle unit)

shifted (2nd / f / SHIFT layer) (press #1 first): $\cos^{-1}$ — Inverse cosine $\cos^{-1}$

Keyboard: c

#12 tan Functions

Primary: Tangent (current angle unit)

shifted (2nd / f / SHIFT layer) (press #1 first): $\tan^{-1}$ — Inverse tangent $\tan^{-1}$

Keyboard: t

#13 +/− Digits & number entry

Primary: Change sign (+/−) of the number being typed or the current result

shifted (2nd / f / SHIFT layer) (press #1 first): $\sqrt[3]{}$ — Cube root $\sqrt[3]{}$

#14 ▶ Clear, delete & power

Primary: Backspace — delete the last digit typed

shifted (2nd / f / SHIFT layer) (press #1 first): x^3 — Cube x^3

Keyboard: Backspace Delete

#15 [(Operators & execute

Primary: Opening bracket (

shifted (2nd / f / SHIFT layer) (press #1 first): X ↔ Y — x≷y — exchange X and Y

Keyboard: (

#16)] Operators & execute

Primary: Closing bracket). Closing brackets at the end of a line may be omitted

shifted (2nd / f / SHIFT layer) (press #1 first): 1/x — Reciprocal x^{-1} (1/x)

Keyboard:)

#17 xy Operators & execute

Primary: Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)

shifted (2nd / f / SHIFT layer) (press #1 first): x1/y — x-th root of y

Keyboard: ^

#18 MR Memory, variables & registers

Primary: MR — memory recall

shifted (2nd / f / SHIFT layer) (press #1 first): Min — Min — store into memory (overwrite)

#19 7 Digits & number entry

Primary: Digit 7

shifted (2nd / f / SHIFT layer) (press #1 first): $\bar{x}$ — $\bar{x}$ — mean

Keyboard: 7

#20 8 Digits & number entry

Primary: Digit 8

shifted (2nd / f / SHIFT layer) (press #1 first): σn — σ — population standard deviation

Keyboard: 8

#21 9 Digits & number entry

Primary: Digit 9

shifted (2nd / f / SHIFT layer) (press #1 first): σn−1 — Sx — sample standard deviation

Keyboard: 9

#22 C Clear, delete & power

Primary: C — clear the entry

shifted (2nd / f / SHIFT layer) (press #1 first): X ↔ M — X ↔ M — swap display and memory

#23 AC Clear, delete & power

Primary: AC — clear all of the current calculation

shifted (2nd / f / SHIFT layer) (press #1 first): SAC — Scl — clear statistics data

Keyboard: Escape

#24 4 Digits & number entry

Primary: Digit 4

shifted (2nd / f / SHIFT layer) (press #1 first): Σx^2 — Σx^2 — sum of squares

Keyboard: 4

#25 5 Digits & number entry

Primary: Digit 5

shifted (2nd / f / SHIFT layer) (press #1 first): Σx — Σx — sum of x

Keyboard: 5

#26 6 Digits & number entry

Primary: Digit 6

shifted (2nd / f / SHIFT layer) (press #1 first): n — n — count of data

Keyboard: 6

#27 × Operators & execute

Primary: Multiplication ×

shifted (2nd / f / SHIFT layer) (press #1 first): ENG← — ←ENG — engineering notation, shifting the other way

Keyboard: * x

#28 ÷ `Operators & execute`

Primary: Division ÷

shifted (2nd / f / SHIFT layer) (press #1 first): ENG — ENG — engineering notation (exponent multiple of 3)

Keyboard: /

#29 1 `Digits & number entry`

Primary: Digit 1

shifted (2nd / f / SHIFT layer) (press #1 first): nPr — Permutations nPr

Keyboard: 1

#30 2 `Digits & number entry`

Primary: Digit 2

shifted (2nd / f / SHIFT layer) (press #1 first): nCr — Combinations nCr

Keyboard: 2

#31 3 `Digits & number entry`

Primary: Digit 3

shifted (2nd / f / SHIFT layer) (press #1 first): x! — Factorial n! (also on non-integers via the gamma function where supported)

Keyboard: 3

#32 + Operators & execute

Primary: Addition +

shifted (2nd / f / SHIFT layer) (press #1 first): R→P — R→P — rectangular to polar

Keyboard: +

#33 – Operators & execute

Primary: Subtraction –

shifted (2nd / f / SHIFT layer) (press #1 first): P→R — P→R — polar to rectangular

Keyboard: -

#34 0 Digits & number entry

Primary: Digit 0

shifted (2nd / f / SHIFT layer) (press #1 first): RND — Round to the displayed precision

Keyboard: 0

#35 • Digits & number entry

Primary: Decimal point. Pressing it first gives "0."; a second point in the same number is ignored

shifted (2nd / f / SHIFT layer) (press #1 first): RAN# — Random number 0–1

Keyboard: . ,

#36 EXP `Digits & number entry`

Primary: EXP — enter a power-of-ten exponent (scientific notation)

shifted (2nd / f / SHIFT layer) (press #1 first): π — π (3.14159265…) as an exact constant

#37 = `Operators & execute`

Primary: Equals — complete the calculation

shifted (2nd / f / SHIFT layer) (press #1 first): % — Percent %

Keyboard: [=]

#38 M+ `Memory, variables & registers`

Primary: M+ — add the value to independent memory M

shifted (2nd / f / SHIFT layer) (press #1 first): M− — M− — subtract the value from memory M

5 · Worked examples

23 examples, each run on this calculator's keys by the U-45 checker. Follow the chips in order.

Example 5.1 — Order of operations

Calculate 12 + 3 × 4. Press Reset first.

#29 1 → #30 2 → #32 + → #31 3 → #27 × → #24 4 → #37 =

Display shows 24. **— expected 24. ✔ verified on the app.**

Multiplication is done before addition, so the calculator holds the + until the × part is finished.

Example 5.2 — Brackets first

Calculate (12 + 8) ÷ 4. Press Reset first.

#15 [(→ #29 1 → #30 2 → #32 + → #20 8 → #16)] → #28 ÷ → #24 4 → #37 =

Display shows 5. **— expected 5. ✔ verified on the app.**

Brackets force the sum to be worked out before the division.

Example 5.3 — Two brackets multiplied

Calculate (3 + 4) × (9 − 2). Press Reset first.

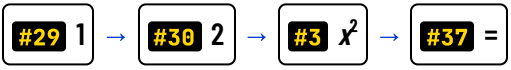


Display shows 49. **— expected 49. ✔ verified on the app.**

Both brackets are evaluated first, then multiplied.

Example 5.4 — Squares

Calculate 12^2. Press Reset first.



Display shows 144. **— expected 144. ✔ verified on the app.**

x^2 squares the value — quicker and safer than typing the number twice.

Example 5.5 — Square roots

Calculate √144. Press Reset first.



Display shows 12. **— expected 12. ✔ verified on the app.**

The square-root key returns the principal (positive) root.

Example 5.6 — Powers

Calculate **2^10**. Press **Reset** first.

#30 2 → **#17** x^y → **#29** 1 → **#34** 0 → **#37** =

Display shows `1024.` — expected 1024. ✔ verified on the app.

The power key raises the first number to the second.

Example 5.7 — Reciprocals

Calculate **1 ÷ 4 (x^{-1})**. Press **Reset** first.

#24 4 → **#1** SHIFT → **#16** 1/*x* → **#37** =

Display shows `0.25` — expected 0.25. ✔ verified on the app.

x^{-1} (1/x) gives the reciprocal — handy for resistors in parallel or rates.

Example 5.8 — Trigonometry (degrees)

Calculate **tan 45° (Deg mode)**. Press **Reset** first.

#24 4 → **#25** 5 → **#12** tan → **#37** =

Display shows `1.` — expected 1. ✔ verified on the app.

Trig functions use the current angle unit; these answers assume degrees (the start-up setting).

Example 5.9 — Common logarithm

Calculate **log 1000**. Press **Reset** first.

#29 1 → #34 0 → #34 0 → #34 0 → #4 log → #37 =

Display shows 3. — expected 3. ✔ verified on the app.

log is base-10: it counts the zeros of a power of ten.

Example 5.10 — Natural logarithm

Calculate **ln 1**. Press **Reset** first.

#29 1 → #5 ln → #37 =

Display shows 0. — expected 0. ✔ verified on the app.

ln is the natural (base-e) logarithm.

Example 5.11 — Antilog 10^x

Calculate **10^2**. Press **Reset** first.

#30 2 → #1 SHIFT → #4 10^x → #37 =

Display shows 100. — expected 100. ✔ verified on the app.

10^x is the inverse of log.

Example 5.12 — Exponential e^x

Calculate e^1. Press **Reset** first.

#29 1 → **#1** SHIFT → **#5** e^x → **#37** =

Display shows `2.718281828` — expected 2.71828183. ✔ verified on the app.

e^x is the inverse of ln.

Example 5.13 — Negative numbers

Calculate −5 + 12. Press **Reset** first.

#25 5 → **#13** +/− → **#32** + → **#29** 1 → **#30** 2 → **#37** =

Display shows `7.` — expected 7. ✔ verified on the app.

Use the sign key for a negative number — not the subtraction key.

Example 5.14 — Scientific notation

Calculate 6×10^2 × 3. Press **Reset** first.

#26 6 → **#36** EXP → **#30** 2 → **#27** × → **#31** 3 → **#37** =

Display shows `1800.` — expected 1800. ✔ verified on the app.

The exponent key enters $\times 10^n$ without typing × 10 ^.

Example 5.15 — Area of a circle πr²

Calculate π × 3². Press **Reset** first.

#1 SHIFT → **#36** π → **#27** × → **#31** 3 → **#3** x^2 → **#37** =

Display shows `28.27433388` — expected 28.27433388. ✔ verified on the app.

π is stored to full precision — never type 3.14.

Example 5.16 — Combinations nCr

Calculate 5C2. Press **Reset** first.

#25 5 → **#1** SHIFT → **#30** *nCr* → **#30** 2 → **#37** =

Display shows `10.` — expected 10. ✔ verified on the app.

nCr counts selections where order does not matter.

Example 5.17 — Factorials

Calculate 5!. Press **Reset** first.

#25 5 → **#1** SHIFT → **#31** *x!* → **#37** =

Display shows `120.` — expected 120. ✔ verified on the app.

n! multiplies 1×2×…×n; 0! is 1 by definition.

Example 5.18 — Continue from the answer

Calculate 100 × 4 = , then Ans ÷ 2. Press Reset first.

#29 1 → #34 0 → #34 0 → #27 × → #24 4 → #37 = → #28 ÷ → #30 2 → #37 =

Display shows 200. **— expected 200. ✔ verified on the app.**

Pressing an operator straight after a result continues from that answer (Ans).

Example 5.19 — Percent of an amount

Calculate 15 % of 80. Press Reset first.

#20 8 → #34 0 → #27 × → #29 1 → #25 5 → #1 SHIFT → #37 % → #37 =

Display shows 12. **— expected 12. ✔ verified on the app.**

The percent key turns the second number into a percentage of the first.

Example 5.20 — Average of three numbers

Calculate (4 + 8 + 15) ÷ 3. Press Reset first.

#15 [(→ #24 4 → #32 + → #20 8 → #32 + → #29 1 → #25 5 → #16)] → #28 ÷ → #31 3 → #37 =

Display shows 9. **— expected 9. ✔ verified on the app.**

Add up the values, then divide by how many there are.

Example 5.21 — Mixed operations

Calculate 50 − 4 × 6 + 7. Press Reset first.

#25 5 → #34 0 → #33 − → #24 4 → #27 × → #26 6 → #32 + → #19 7 → #37 =

Display shows 33. **— expected 33. ✔ verified on the app.**

Only the × part is done first; the − and + are then worked left to right.

Example 5.22 — Dividing twice

Calculate 360 ÷ 4 ÷ 9. Press Reset first.

#31 3 → #26 6 → #34 0 → #28 ÷ → #24 4 → #28 ÷ → #21 9 → #37 =

Display shows 10. **— expected 10. ✔ verified on the app.**

Division is worked left to right: the first quotient is divided again.

Example 5.23 — Brackets inside a longer formula

Calculate ((3 + 5) × 4 − 2) ÷ 10. Press Reset first.

#15 [(→ #15 [(→ #31 3 → #32 + → #25 5 → #16)] → #27 × → #24 4 → #33 − → #30 2 → #16)] → #28 ÷ → #29 1 → #34 0 → #37 =

Display shows 3. **— expected 3. ✔ verified on the app.**

The innermost brackets are worked out first, then the outer brackets, then the division.

6 · Tricks & hacks

Hack 1: Order of operations

Algebraic one-line models respect × ÷ before + −, so 2 + 3 × 4 = 14 without brackets.

Hack 2: Constant calculations

Repeat the last operation by pressing = again (where supported) — handy for repeated discounts or growth.

Hack 3: Memory

Use the memory keys to hold a subtotal while you calculate something else.

Hack 4: Sign change

Use +/− after typing the number to make it negative.

Hack 5: Statistics

Enter each data value with Σ+/DATA; then read mean and deviation with the shifted keys.

Hack 6: Hidden functions on this model

34 extra functions sit on shifted layers. Press the layer key once, then the key: for example #1 then #3 → √; #1 then #4 → 10x; #1 then #5 → ex; #1 then #7 → d/c.

Hack 7: Find any key by number

Use Appendix B (function finder): look up the function, read its key number, then find that number on the keypad. Numbers run left → right, top → bottom from #1 to #38.

Hack 8: Type faster on a computer

27 keys on this model have a keyboard shortcut (see chapter 7). Typing digits and operators is much faster than clicking.

Hack 9: Check before you trust

Estimate first (round each number) and compare with the display. If the answer is 10× or 100× off, a decimal point or percent key was missed.

Hack 10: Copy results out

Press Copy in the bar above the calculator to put the display text on the clipboard instead of retyping it.

Hack 11: Colour blind or low vision?

Hi-Contrast colours use black/white/yellow/blue/green/red with black or white text for maximum contrast; key numbers stay on in both colour schemes.

7 · Keyboard shortcuts

With the calculator open on a computer, these physical keys press the numbered keys below (Ctrl/Alt/⌘ combinations are left to the browser). Shifted functions are reached by clicking the shift key first.

Keyboard	#	Key	Does
q	#3	x^2	Square x^2
l	#4	log	Common logarithm $\log_{10}$
n	#5	ln	Natural logarithm ln
s	#10	sin	Sine (uses the current angle unit)
c	#11	cos	Cosine (current angle unit)
t	#12	tan	Tangent (current angle unit)
Backspace Delete	#14	►	Backspace — delete the last digit typed
(	#15	[(	Opening bracket (
)	#16	)]	Closing bracket). Closing brackets at the end of a line may be omitted
^	#17	xy	Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)
7	#19	7	Digit 7
8	#20	8	Digit 8
9	#21	9	Digit 9
Escape	#23	AC	AC — clear all of the current calculation
4	#24	4	Digit 4
5	#25	5	Digit 5
6	#26	6	Digit 6
* x	#27	×	Multiplication ×
/	#28	÷	Division ÷
1	#29	1	Digit 1

2	#30	2	Digit 2
3	#31	3	Digit 3
+	#32	+	Addition +
-	#33	–	Subtraction –
0	#34	0	Digit 0
. ,	#35	•	Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored
=	#37	=	Equals — complete the calculation

App shortcuts

“# On / # Off”	show / hide key numbers
“Classic / Hi-Contrast”	switch key colours
“Copy”	copy display text
“Fit / 1:1”	scale to screen / design size
“Reset”	restart this calculator
Jump list	switch to another of the 45 calculators

8 · Common mistakes

Mistake 1: Typing +/− first

On one-line models +/− is pressed after the digits: 5 +/− gives −5.

Mistake 2: CE vs C

CE clears only the last entry; C/AC clears the whole calculation.

Mistake 3: Forgetting the angle unit

Check DEG/RAD/GRAD before trig functions.

Mistake 4: Functions after the number

One-line models apply functions to the number already on display: 9 √ = 3 (not √ 9).

Mistake 5: Forgetting Reset between problems

Old values (memory, stack, open brackets, worksheet values) can leak into the next calculation. Press Reset or clear before starting a new problem.

Mistake 6: Pressing the wrong layer

A shift / 2nd indicator that is still lit will turn the next key into its shifted function. Look at the display indicators before pressing.

9 · Practice drills

Set A — Calculation drills (124)

Work each one on calculator #23, then check your keys and the answer in Appendix A. Every answer was produced by this calculator.

1. 25 + 5 × 6 = _____
2. 7 + 8 × 9 = _____
3. 150 + 12 × 11 = _____
4. 3 + 14 × 15 = _____
5. 48 + 6 × 7 = _____
6. 1000 + 25 × 4 = _____
7. 19 + 21 × 3 = _____
8. 64 + 8 × 8 = _____
9. (45 + 15) ÷ 6 = _____
10. (9 + 7) ÷ 2 = _____
11. (100 + 60) ÷ 8 = _____
12. (33 + 17) ÷ 5 = _____
13. (1 + 2) ÷ 4 = _____
14. (250 + 350) ÷ 12 = _____
15. (5 + 6) × (10 − 3) = _____
16. (12 + 3) × (8 − 5) = _____

17. (1.5 + 2.5) × (7 − 4) = _____

18. (20 + 30) × (15 − 5) = _____

19. (9 + 1) × (100 − 1) = _____

20. (11 + 13) × (17 − 9) = _____

21. 25^2 = _____

22. 7^2 = _____

23. 1.5^2 = _____

24. 19^2 = _____

25. 33^2 = _____

26. 0.5^2 = _____

27. 125^2 = _____

28. 64^2 = _____

29. $\sqrt{625}$ = _____

30. $\sqrt{2}$ = _____

31. $\sqrt{50}$ = _____

32. $\sqrt{1024}$ = _____

33. $\sqrt{3}$ = _____

34. $\sqrt{9801}$ = _____

35. 3^5 = _____

36. 1.05^3 = _____

37. 5^4 = _____

38. 7^3 = _____

39. 1.5^2 = _____

40. 10^6 = _____

41. 2^0.5 = _____

42. 1 ÷ 8 (x^{-1}) = _____

43. 1 ÷ 0.25 (x^{-1}) = _____

44. 1 ÷ 16 (x^{-1}) = _____

45. 1 ÷ 40 (x^{-1}) = _____

46. 1 ÷ 0.2 (x^{-1}) = _____

47. sin 90° (Deg mode) = _____

48. cos 0° (Deg mode) = _____

49. log 100 = _____

50. log 10000 = _____

51. ln 10 = _____

52. ln 2 = _____

53. 10^3 = _____

54. e^2 = _____

55. −20 + 7 = _____

56. −3 + 3 = _____

57. −100 + 250 = _____

58. −8 + 1 = _____

59. 2×10^3 × 4 = _____

60. 1.5×10^4 × 2 = _____

61. 3×10^5 × 3 = _____

62. 4.5×10^2 × 2 = _____

63. π × 10^2 = _____

64. π × 2.5^2 = _____

65. π × 7^2 = _____

66. π × 1.2^2 = _____

67. 10C3 = _____

68. 6C6 = _____

69. 6! = _____

70. 0! = _____

71. 15 × 6 = , then Ans ÷ 9 = _____

72. 7.5 × 8 = , then Ans ÷ 3 = _____

73. 250 × 12 = , then Ans ÷ 100 = _____

74. 8 % of 250 = _____

75. 25 % of 360 = _____

76. 12.5 % of 40 = _____

77. 10 % of 450 = _____

78. 20 % of 125 = _____

79. 5 % of 640 = _____

80. 75 % of 48 = _____

81. 30 % of 90 = _____

82. (12 + 15 + 18) ÷ 3 = _____

83. (70 + 85 + 91) ÷ 3 = _____

84. (2.5 + 3.5 + 6) ÷ 3 = _____

85. (100 + 200 + 600) ÷ 3 = _____

86. 100 − 3 × 12 + 9 = _____

87. 18 − 2 × 5 + 40 = _____

88. 7.5 − 1.5 × 3 + 2 = _____

89. 64 − 8 × 4 + 10 = _____

90. 300 − 25 × 4 + 55 = _____

91. 9 − 3 × 3 + 1 = _____

92. 45 − 5 × 5 + 15 = _____

93. 1000 ÷ 8 ÷ 5 = _____

94. 96 ÷ 2 ÷ 3 = _____

95. 7200 ÷ 60 ÷ 60 = _____

96. 500 ÷ 4 ÷ 25 = _____

97. 81 ÷ 3 ÷ 3 = _____

98. ((10 + 2) × 3 − 6) ÷ 6 = _____

99. ((1.5 + 2.5) × 6 − 4) ÷ 5 = _____

100. ((20 + 30) × 2 − 25) ÷ 3 = _____

101. ((7 + 8) × 9 − 15) ÷ 12 = _____

102. 12 + 3 × 4 = _____

103. (12 + 8) ÷ 4 = _____

104. (3 + 4) × (9 − 2) = _____

105. 12^2 = _____

106. $\sqrt{144}$ = _____

107. 2^10 = _____

108. 1 ÷ 4 (x^{-1}) = _____

109. tan 45° (Deg mode) = _____

110. log 1000 = _____

111. ln 1 = _____

112. 10^2 = _____

113. e^1 = _____

114. −5 + 12 = _____

115. 6×10^2 × 3 = _____

116. π × 3^2 = _____

117. 5C2 = _____

118. 5! = _____

119. 100 × 4 = , then Ans ÷ 2 = _____

120. 15 % of 80 = _____

121. (4 + 8 + 15) ÷ 3 = _____

122. 50 − 4 × 6 + 7 = _____

123. 360 ÷ 4 ÷ 9 = _____

124. ((3 + 5) × 4 − 2) ÷ 10 = _____

Set B — Key-finding drills (62)

Without looking at the table, find each key on the picture. Answers in Appendix A.

1. Which key is “SHIFT” (SHIFT / 2nd)? _____
2. Which key is “MODE” (MODE)? _____
3. Which key is “x^2” (Square x^2)? _____
4. Which key is “log” (Common logarithm $\log_{10}$)? _____
5. Which key is “ln” (Natural logarithm ln)? _____
6. Which key is “ON” (ON)? _____
7. Which key is “ab/c” (a b/c)? _____
8. Which key is “°′ ″” (Degrees-minutes-seconds entry/display °′ ″)? _____
9. Which key is “hyp” (Hyperbolic prefix)? _____
10. Which key is “sin” (Sine)? _____

11. Which key is “cos” (Cosine)? _____

12. Which key is “tan” (Tangent)? _____

13. Which key is “+/–” (Change sign)? _____

14. Which key is “ ► ” (Backspace)? _____

15. Which key is “[(” (Opening bracket)? _____

16. Which key is “)]” (Closing bracket). Closing brackets at the end of a line may be omitted)? _____

17. Which key is “xy” (Power x^y / ^)? _____

18. Which key is “MR” (MR)? _____

19. Which key is “C” (C)? _____

20. Which key is “AC” (AC)? _____

21. Which key is “×” (Multiplication ×)? _____

22. Which key is “÷” (Division ÷)? _____

23. Which key is “+” (Addition +)? _____

24. Which key is “–” (Subtraction –)? _____

25. Which key is “•” (Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored)? _____

26. Which key is “EXP” (EXP)? _____

27. Which key is “=” (Equals)? _____

28. Which key is “M+” (M+)? _____

29. How do you reach “√”? _____

30. How do you reach “10x”? _____

31. How do you reach “ex”? _____

32. How do you reach “d/c”? _____

33. How do you reach “ ← ”? _____

34. How do you reach “$\sin^{-1}$”? _____

35. How do you reach “$\cos^{-1}$”? _____

36. How do you reach “$\tan^{-1}$”? _____

37. How do you reach “$\sqrt[3]{}$”? _____

38. How do you reach “x^3”? _____

39. How do you reach “X ↔ Y”? _____

40. How do you reach “1/x”? _____

41. How do you reach “x1/y”? _____

42. How do you reach “Min”? _____

43. How do you reach “$\bar{x}$”? _____

44. How do you reach “σn”? _____

45. How do you reach “σn−1”? _____

46. How do you reach “X ↔ M”? _____

47. How do you reach “SAC”? _____

48. How do you reach “Σx^2”? _____

49. How do you reach “Σx”? _____

50. How do you reach “n”? _____

51. How do you reach “ENG←”? _____

52. How do you reach “ENG”? _____

53. How do you reach “nPr”? _____

54. How do you reach “nCr”? _____

55. How do you reach “x!”? _____

56. How do you reach “R→P”? _____

57. How do you reach “P→R”? _____

58. How do you reach “RND”? _____

59. How do you reach “RAN#”? _____

60. How do you reach “π”? _____

61. How do you reach “%”? _____

62. How do you reach “M−”? _____

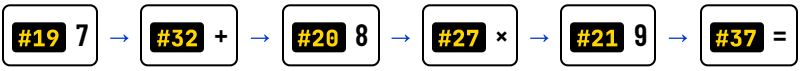
Appendix A — Drill answers

Set A

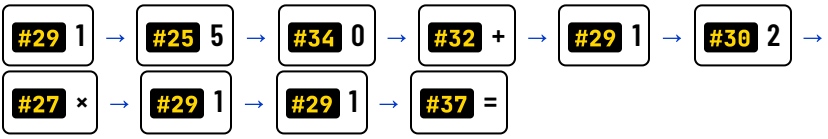
1. 55 (display `55.`)



2. 79 (display `79.`)



3. 282 (display `282.`)



4. 213 (display `213.`)



5. 90 (display `90.`)



6. 1100 (display `1100.`)



7. 82 (display `82.`)

`#29` 1 → `#21` 9 → `#32` + → `#30` 2 → `#29` 1 → `#27` × → `#31` 3 → `#37` =

8. 128 (display `128.`)

`#26` 6 → `#24` 4 → `#32` + → `#20` 8 → `#27` × → `#20` 8 → `#37` =

9. 10 (display `10.`)

`#15` [(→ `#24` 4 → `#25` 5 → `#32` + → `#29` 1 → `#25` 5 → `#16`)] → `#28` ÷ → `#26` 6 → `#37` =

10. 8 (display `8.`)

`#15` [(→ `#21` 9 → `#32` + → `#19` 7 → `#16`)] → `#28` ÷ → `#30` 2 → `#37` =

11. 20 (display `20.`)

`#15` [(→ `#29` 1 → `#34` 0 → `#34` 0 → `#32` + → `#26` 6 → `#34` 0 → `#16`)] → `#28` ÷ → `#20` 8 → `#37` =

12. 10 (display `10.`)

`#15` [(→ `#31` 3 → `#31` 3 → `#32` + → `#29` 1 → `#19` 7 → `#16`)] → `#28` ÷ → `#25` 5 → `#37` =

13. 0.75 (display `0.75`)

`#15` [(→ `#29` 1 → `#32` + → `#30` 2 → `#16`)] → `#28` ÷ → `#24` 4 → `#37` =

14. 50 (display 50.)

#15 [(→ #30 2 → #25 5 → #34 0 → #32 + → #31 3 → #25 5 → #34 0 → #16)] → #28 ÷ → #29 1 → #30 2 → #37 =

15. 77 (display 77.)

#15 [(→ #25 5 → #32 + → #26 6 → #16)] → #27 × → #15 [(→ #29 1 → #34 0 → #33 − → #31 3 → #16)] → #37 =

16. 45 (display 45.)

#15 [(→ #29 1 → #30 2 → #32 + → #31 3 → #16)] → #27 × → #15 [(→ #20 8 → #33 − → #25 5 → #16)] → #37 =

17. 12 (display 12.)

#15 [(→ #29 1 → #35 • → #25 5 → #32 + → #30 2 → #35 • → #25 5 → #16)] → #27 × → #15 [(→ #19 7 → #33 − → #24 4 → #16)] → #37 =

18. 500 (display 500.)

#15 [(→ #30 2 → #34 0 → #32 + → #31 3 → #34 0 → #16)] → #27 × → #15 [(→ #29 1 → #25 5 → #33 − → #25 5 → #16)] → #37 =

19. 990 (display `990.`)

`#15` [(→ `#21` 9 → `#32` + → `#29` 1 → `#16`)] → `#27` × →
`#15` [(→ `#29` 1 → `#34` 0 → `#34` 0 → `#33` − → `#29` 1 →
`#16`)] → `#37` =

20. 192 (display `192.`)

`#15` [(→ `#29` 1 → `#29` 1 → `#32` + → `#29` 1 → `#31` 3 →
`#16`)] → `#27` × → `#15` [(→ `#29` 1 → `#19` 7 → `#33` − →
`#21` 9 → `#16`)] → `#37` =

21. 625 (display `625.`)

`#30` 2 → `#25` 5 → `#3` x^2 → `#37` =

22. 49 (display `49.`)

`#19` 7 → `#3` x^2 → `#37` =

23. 2.25 (display `2.25`)

`#29` 1 → `#35` • → `#25` 5 → `#3` x^2 → `#37` =

24. 361 (display `361.`)

`#29` 1 → `#21` 9 → `#3` x^2 → `#37` =

25. 1089 (display `1089.`)

`#31` 3 → `#31` 3 → `#3` x^2 → `#37` =

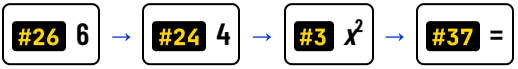
26. 0.25 (display `0.25`)

`#34` 0 → `#35` • → `#25` 5 → `#3` x^2 → `#37` =

27. 15625 (display `15625.`)



28. 4096 (display `4096.`)



29. 25 (display `25.`)



30. 1.41421356 (display `1.414213562`)



31. 7.07106781 (display `7.071067812`)



32. 32 (display `32.`)



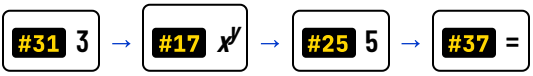
33. 1.73205081 (display `1.732050808`)



34. 99 (display `99.`)



35. 243 (display `243.`)



36. 1.157625 (display `1.157625`)

#29 1 → #35 • → #34 0 → #25 5 → #17 x^y → #31 3 → #37 =

37. 625 (display `625.`)

#25 5 → #17 x^y → #24 4 → #37 =

38. 343 (display `343.`)

#19 7 → #17 x^y → #31 3 → #37 =

39. 2.25 (display `2.25`)

#29 1 → #35 • → #25 5 → #17 x^y → #30 2 → #37 =

40. 1000000.000 (display `1000000.`)

#29 1 → #34 0 → #17 x^y → #26 6 → #37 =

41. 1.41421356 (display `1.414213562`)

#30 2 → #17 x^y → #34 0 → #35 • → #25 5 → #37 =

42. 0.125 (display `0.125`)

#20 8 → #1 SHIFT → #16 $1/x$ → #37 =

43. 4 (display `4.`)

#34 0 → #35 • → #30 2 → #25 5 → #1 SHIFT → #16 $1/x$ → #37 =

44. 0.0625 (display `0.0625`)

#29 1 → #26 6 → #1 SHIFT → #16 $1/x$ → #37 =

45. 0.025 (display `0.025`)

`#24` 4 → `#34` 0 → `#1` SHIFT → `#16` 1/*x* → `#37` =

46. 5 (display `5.`)

`#34` 0 → `#35` • → `#30` 2 → `#1` SHIFT → `#16` 1/*x* → `#37` =

47. 1 (display `1.`)

`#21` 9 → `#34` 0 → `#10` sin → `#37` =

48. 1 (display `1.`)

`#34` 0 → `#11` cos → `#37` =

49. 2 (display `2.`)

`#29` 1 → `#34` 0 → `#34` 0 → `#4` log → `#37` =

50. 4 (display `4.`)

`#29` 1 → `#34` 0 → `#34` 0 → `#34` 0 → `#34` 0 → `#4` log → `#37` =

51. 2.30258509 (display `2.302585093`)

`#29` 1 → `#34` 0 → `#5` ln → `#37` =

52. 0.69314718 (display `0.6931471806`)

`#30` 2 → `#5` ln → `#37` =

53. 1000 (display `1000.`)

`#31` 3 → `#1` SHIFT → `#4` 10^x → `#37` =

54. 7.3890561 (display `7.389056099`)

`#30` 2 → `#1` SHIFT → `#5` e^x → `#37` =

55. -13 (display `-13.`)

`#30` 2 → `#34` 0 → `#13` +/− → `#32` + → `#19` 7 → `#37` =

56. 0 (display `0.`)

`#31` 3 → `#13` +/− → `#32` + → `#31` 3 → `#37` =

57. 150 (display `150.`)

`#29` 1 → `#34` 0 → `#34` 0 → `#13` +/− → `#32` + → `#30` 2 → `#25` 5 → `#34` 0 → `#37` =

58. -7 (display `-7.`)

`#20` 8 → `#13` +/− → `#32` + → `#29` 1 → `#37` =

59. 8000 (display `8000.`)

`#30` 2 → `#36` EXP → `#31` 3 → `#27` × → `#24` 4 → `#37` =

60. 30000 (display `30000.`)

`#29` 1 → `#35` • → `#25` 5 → `#36` EXP → `#24` 4 → `#27` × → `#30` 2 → `#37` =

61. 900000 (display `900000.`)

`#31` 3 → `#36` EXP → `#25` 5 → `#27` × → `#31` 3 → `#37` =

62. 900 (display `900.`)

`#24` 4 → `#35` • → `#25` 5 → `#36` EXP → `#30` 2 → `#27` × → `#30` 2 → `#37` =

63. 314.15926536 (display `314.1592654`)

#1 SHIFT → #36 π → #27 × → #29 1 → #34 0 → #3 x^2 → #37 =

64. 19.63495408 (display `19.63495408`)

#1 SHIFT → #36 π → #27 × → #30 2 → #35 • → #25 5 → #3 x^2 → #37 =

65. 153.93804003 (display `153.93804`)

#1 SHIFT → #36 π → #27 × → #19 7 → #3 x^2 → #37 =

66. 4.52389342 (display `4.523893421`)

#1 SHIFT → #36 π → #27 × → #29 1 → #35 • → #30 2 → #3 x^2 → #37 =

67. 120 (display `120.`)

#29 1 → #34 0 → #1 SHIFT → #30 *n*C*r* → #31 3 → #37 =

68. 1 (display `1.`)

#26 6 → #1 SHIFT → #30 *n*C*r* → #26 6 → #37 =

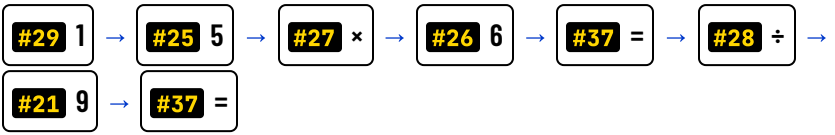
69. 720 (display `720.`)

#26 6 → #1 SHIFT → #31 *x*! → #37 =

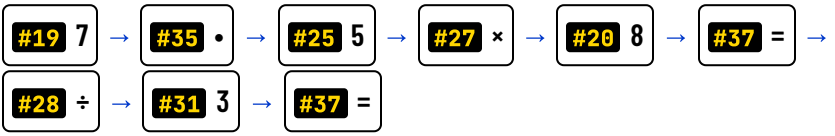
70. 1 (display `1.`)

#34 0 → #1 SHIFT → #31 *x*! → #37 =

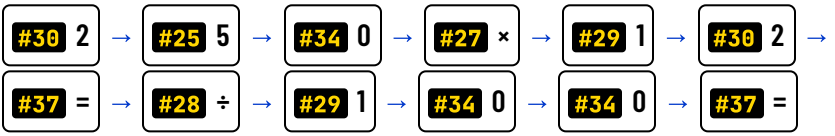
71. 10 (display `10.`)



72. 20 (display `20.`)



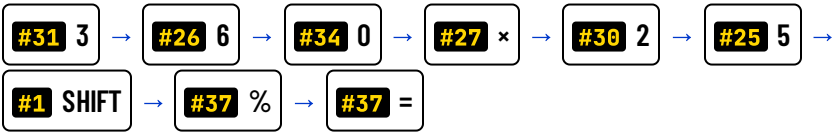
73. 30 (display `30.`)



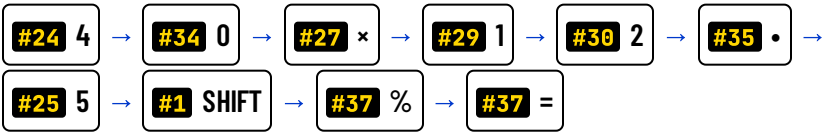
74. 20 (display `20.`)



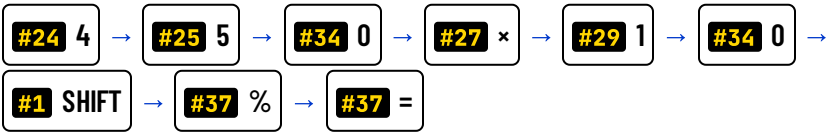
75. 90 (display `90.`)



76. 5 (display `5.`)



77. 45 (display `45.`)



78. 25 (display `25.`)

`#29` 1 → `#30` 2 → `#25` 5 → `#27` × → `#30` 2 → `#34` 0 → `#1` SHIFT → `#37` % → `#37` =

79. 32 (display `32.`)

`#26` 6 → `#24` 4 → `#34` 0 → `#27` × → `#25` 5 → `#1` SHIFT → `#37` % → `#37` =

80. 36 (display `36.`)

`#24` 4 → `#20` 8 → `#27` × → `#19` 7 → `#25` 5 → `#1` SHIFT → `#37` % → `#37` =

81. 27 (display `27.`)

`#21` 9 → `#34` 0 → `#27` × → `#31` 3 → `#34` 0 → `#1` SHIFT → `#37` % → `#37` =

82. 15 (display `15.`)

`#15` [(→ `#29` 1 → `#30` 2 → `#32` + → `#29` 1 → `#25` 5 → `#32` + → `#29` 1 → `#20` 8 → `#16`)] → `#28` ÷ → `#31` 3 → `#37` =

83. 82 (display `82.`)

`#15` [(→ `#19` 7 → `#34` 0 → `#32` + → `#20` 8 → `#25` 5 → `#32` + → `#21` 9 → `#29` 1 → `#16`)] → `#28` ÷ → `#31` 3 → `#37` =

84. 4 (display `4.`)

`#15` [(→ `#30` 2 → `#35` • → `#25` 5 → `#32` + → `#31` 3 → `#35` • → `#25` 5 → `#32` + → `#26` 6 → `#16`)] → `#28` ÷ → `#31` 3 → `#37` =

85. 300 (display `300.`)

`#15` [(→ `#29` 1 → `#34` 0 → `#34` 0 → `#32` + → `#30` 2 → `#34` 0 → `#34` 0 → `#32` + → `#26` 6 → `#34` 0 → `#34` 0 → `#16`)] → `#28` ÷ → `#31` 3 → `#37` =

86. 73 (display `73.`)

`#29` 1 → `#34` 0 → `#34` 0 → `#33` − → `#31` 3 → `#27` × → `#29` 1 → `#30` 2 → `#32` + → `#21` 9 → `#37` =

87. 48 (display `48.`)

`#29` 1 → `#20` 8 → `#33` − → `#30` 2 → `#27` × → `#25` 5 → `#32` + → `#24` 4 → `#34` 0 → `#37` =

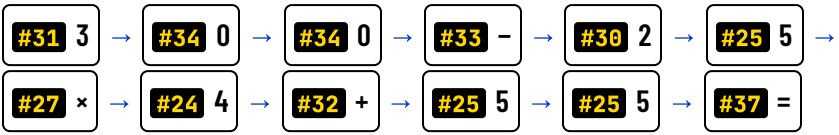
88. 5 (display `5.`)

`#19` 7 → `#35` • → `#25` 5 → `#33` − → `#29` 1 → `#35` • → `#25` 5 → `#27` × → `#31` 3 → `#32` + → `#30` 2 → `#37` =

89. 42 (display `42.`)

`#26` 6 → `#24` 4 → `#33` − → `#20` 8 → `#27` × → `#24` 4 → `#32` + → `#29` 1 → `#34` 0 → `#37` =

90. 255 (display `255.`)



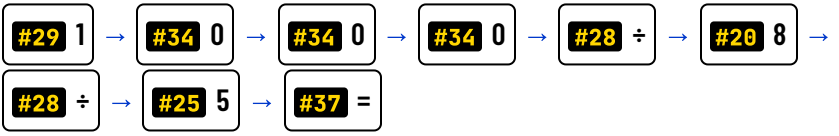
91. 1 (display `1.`)



92. 35 (display `35.`)



93. 25 (display `25.`)



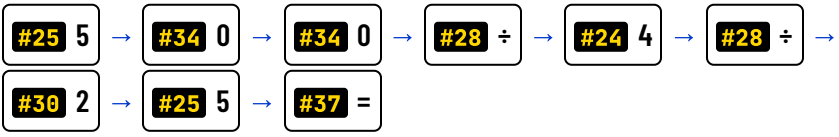
94. 16 (display `16.`)



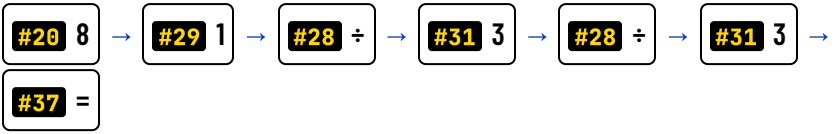
95. 2 (display `2.`)



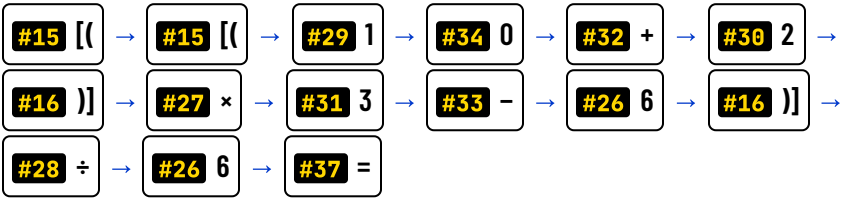
96. 5 (display `5.`)



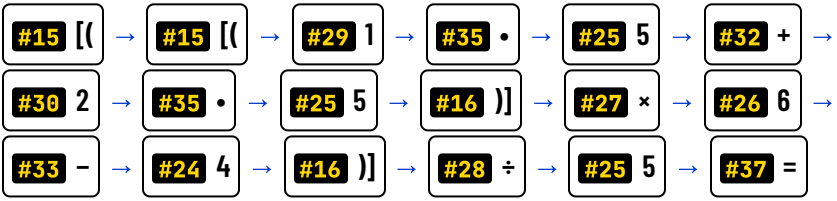
97. 9 (display `9.`)



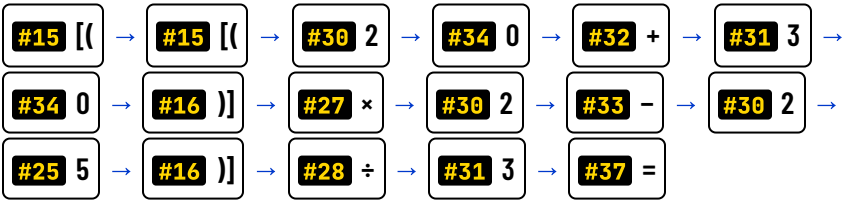
98. 5 (display `5.`)



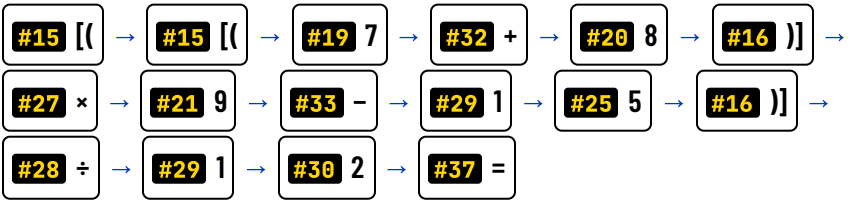
99. 4 (display `4.`)



100. 25 (display `25.`)



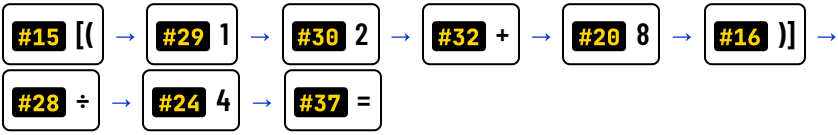
101. 10 (display `10.`)



102. 24 (display `24.`)



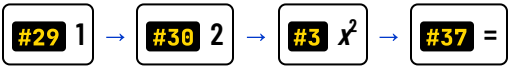
103. 5 (display `5.`)



104. 49 (display `49.`)



105. 144 (display `144.`)



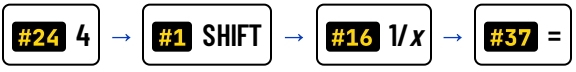
106. 12 (display `12.`)



107. 1024 (display `1024.`)



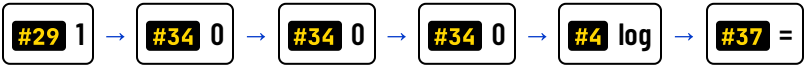
108. 0.25 (display `0.25`)



109. 1 (display `1.`)



110. 3 (display `3.`)



111. 0 (display `0.`)

`#29` 1 → `#5` ln → `#37` =

112. 100 (display `100.`)

`#30` 2 → `#1` SHIFT → `#4` 10^x → `#37` =

113. 2.71828183 (display `2.718281828`)

`#29` 1 → `#1` SHIFT → `#5` e^x → `#37` =

114. 7 (display `7.`)

`#25` 5 → `#13` +/− → `#32` + → `#29` 1 → `#30` 2 → `#37` =

115. 1800 (display `1800.`)

`#26` 6 → `#36` EXP → `#30` 2 → `#27` × → `#31` 3 → `#37` =

116. 28.27433388 (display `28.27433388`)

`#1` SHIFT → `#36` π → `#27` × → `#31` 3 → `#3` x^2 → `#37` =

117. 10 (display `10.`)

`#25` 5 → `#1` SHIFT → `#30` nCr → `#30` 2 → `#37` =

118. 120 (display `120.`)

`#25` 5 → `#1` SHIFT → `#31` $x!$ → `#37` =

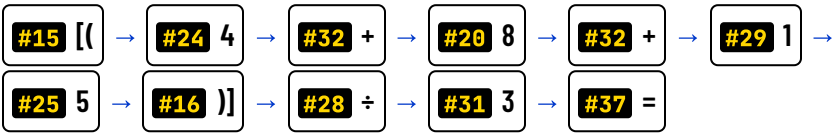
119. 200 (display `200.`)

`#29` 1 → `#34` 0 → `#34` 0 → `#27` × → `#24` 4 → `#37` = →
`#28` ÷ → `#30` 2 → `#37` =

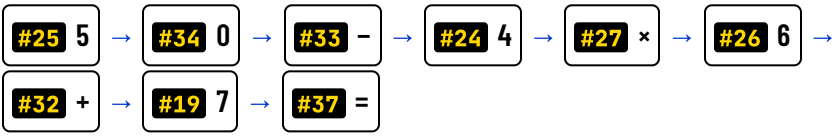
120. 12 (display `12.`)



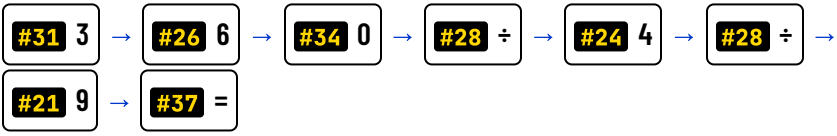
121. 9 (display `9.`)



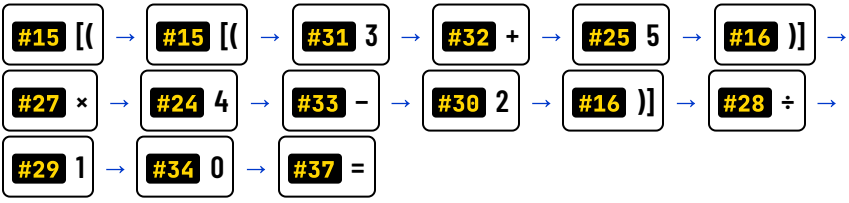
122. 33 (display `33.`)



123. 10 (display `10.`)



124. 3 (display `3.`)



Set B

1. #1
2. #2
3. #3
4. #4
5. #5
6. #6
7. #7
8. #8
9. #9
10. #10
11. #11
12. #12
13. #13
14. #14

15. #15
16. #16
17. #17
18. #18
19. #22
20. #23
21. #27
22. #28
23. #32
24. #33
25. #35
26. #36
27. #37
28. #38
29. #1 SHIFT then #3 x^2
30. #1 SHIFT then #4 log
31. #1 SHIFT then #5 ln
32. #1 SHIFT then #7 ab/c
33. #1 SHIFT then #8 °’”
34. #1 SHIFT then #10 sin
35. #1 SHIFT then #11 cos
36. #1 SHIFT then #12 tan
37. #1 SHIFT then #13 +/-
38. #1 SHIFT then #14 ►
39. #1 SHIFT then #15 [(
40. #1 SHIFT then #16)]
41. #1 SHIFT then #17 xy
42. #1 SHIFT then #18 MR
43. #1 SHIFT then #19 7
44. #1 SHIFT then #20 8
45. #1 SHIFT then #21 9
46. #1 SHIFT then #22 C
47. #1 SHIFT then #23 AC
48. #1 SHIFT then #24 4
49. #1 SHIFT then #25 5
50. #1 SHIFT then #26 6
51. #1 SHIFT then #27 ×
52. #1 SHIFT then #28 ÷
53. #1 SHIFT then #29 1
54. #1 SHIFT then #30 2
55. #1 SHIFT then #31 3
56. #1 SHIFT then #32 +
57. #1 SHIFT then #33 -
58. #1 SHIFT then #34 0
59. #1 SHIFT then #35 •
60. #1 SHIFT then #36 EXP
61. #1 SHIFT then #37 =
62. #1 SHIFT then #38 M+

Appendix B — Function finder (A–Z)

Function	Label	Keys
← ENG — engineering notation, shifting the other way	ENG←	#1 → #27
10^x — antilog	10x	#1 → #4
a b/c — enter fraction / mixed-number parts	ab/c	#7
AC — clear all of the current calculation	AC	#23
Addition +	+	#32
Backspace — delete the last digit typed	►	#14
C — clear the entry	C	#22
Change sign (+/−) of the number being typed or the current result	+/−	#13
Closing bracket). Closing brackets at the end of a line may be omitted	)]	#16
Combinations nCr	nCr	#1 → #30
Common logarithm $\log_{10}$	log	#4
Convert D°M′ S″ to decimal degrees	←	#1 → #8
Cosine (current angle unit)	cos	#11
Cube root $\sqrt[3]{}$	$\sqrt[3]{}$	#1 → #13
Cube x^3	x^3	#1 → #14
d/c — switch between mixed number and improper fraction	d/c	#1 → #7
Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored	•	#35
Degrees-minutes-seconds entry/display °′ ″	°′ ″	#8
Digit 0	0	#34
Digit 1	1	#29
Digit 2	2	#30
Digit 3	3	#31

Digit 4	**4**	**#24**
Digit 5	**5**	**#25**
Digit 6	**6**	**#26**
Digit 7	**7**	**#19**
Digit 8	**8**	**#20**
Digit 9	**9**	**#21**
Division ÷	**÷**	**#28**
ENG — engineering notation (exponent multiple of 3)	**ENG**	**#1 → #28**
Equals — complete the calculation	**=**	**#37**
e^x — exponential	**ex**	**#1 → #5**
EXP — enter a power-of-ten exponent (scientific notation)	**EXP**	**#36**
Factorial n! (also on non-integers via the gamma function where supported)	**x!**	**#1 → #31**
Hyperbolic prefix — the next sin/cos/tan becomes sinh/cosh/tanh	**hyp**	**#9**
Inverse cosine $\cos^{-1}$	**$\cos^{-1}$**	**#1 → #11**
Inverse sine $\sin^{-1}$	**$\sin^{-1}$**	**#1 → #10**
Inverse tangent $\tan^{-1}$	**$\tan^{-1}$**	**#1 → #12**
M+ — add the value to independent memory M	**M+**	**#38**
M− — subtract the value from memory M	**M−**	**#1 → #38**
Min — store into memory (overwrite)	**Min**	**#1 → #18**
MODE — choose calculation mode	**MODE**	**#2**
MR — memory recall	**MR**	**#18**
Multiplication ×	**×**	**#27**
n — count of data	**n**	**#1 → #26**

Natural logarithm ln	ln	#5
ON — power on	ON	#6
Opening bracket (	[(	#15
P → R — polar to rectangular	P → R	#1 → #33
Percent %	%	#1 → #37
Permutations nPr	nPr	#1 → #29
Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)	xy	#17
R → P — rectangular to polar	R → P	#1 → #32
Random number 0–1	RAN#	#1 → #35
Reciprocal x^{-1} (1/x)	1/x	#1 → #16
Round to the displayed precision	RND	#1 → #34
Scl — clear statistics data	SAC	#1 → #23
SHIFT / 2nd — the next key uses its upper (shift-coloured) function	SHIFT	#1
Sine (uses the current angle unit)	sin	#10
Square root √	√	#1 → #3
Square x^2	x^2	#3
Subtraction –	–	#33
Sx — sample standard deviation	σn–1	#1 → #21
Tangent (current angle unit)	tan	#12
$\bar{x}$ — mean	$\bar{x}$	#1 → #19

x-th root of y	**x1/y**	**#1 → #17**
X ↔ M — swap display and memory	**X ↔ M**	**#1 → #22**
x≷y — exchange X and Y	**X ↔ Y**	**#1 → #15**
π (3.14159265…) as an exact constant	**π**	**#1 → #36**
σ — population standard deviation	**σn**	**#1 → #20**
Σx — sum of x	**Σx**	**#1 → #25**
Σx^2 — sum of squares	**Σx^2**	**#1 → #24**

Appendix C — Glossary

Ans
The result of the last calculation, reusable in the next.

AOS
Algebraic Operating System — operator precedence on a one-line display.

RPN
Reverse Polish Notation — operands first, then the operation; no = key.

Stack
The RPN registers X, Y, Z, T where numbers wait for operations.

TVM
Time Value of Money — n, i, PV, PMT, FV.

Natural display
Shows fractions, roots and powers as printed in textbooks.

Shift / 2nd
Prefix key that selects the function printed above a key.

Alpha
Prefix key that types the letter printed beside a key.

Radian
Angle unit; 2π rad = 360°.

Grad
Angle unit; 400 grad = 360°.

Engineering notation
Scientific notation with exponents that are multiples of 3.

Mantissa
The digits of a number in scientific notation.

Memory (M)

An independent register for running totals.

Mark-up

Increase over cost; margin is profit as a percent of selling price.

Grand total (GT)

Sum of all totals since the last clear.

nCr / nPr

Combinations / permutations of n items taken r at a time.

Factorial

n! = 1×2×…×n.

Reciprocal

1/x.

Standard deviation

Spread of data around the mean (σ population, s sample).

Key number

The small badge 1…N on each key in U-45 CALCULATORS, used throughout this manual.

Appendix D — Model notes & limits

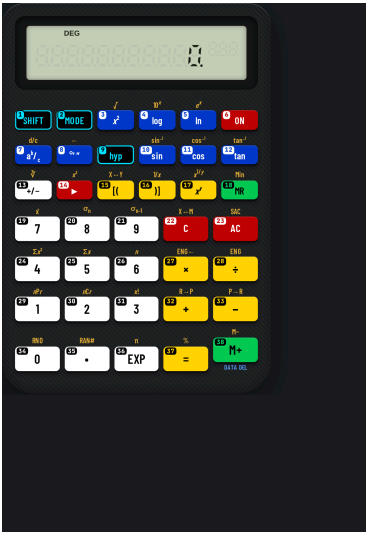
Classic one-line immediate-execution algebraic model. MODE then 0 = COMP, 2 = SD (single-variable stats), 4/5/6 = DEG/RAD/GRA, 7/8/9 = FIX/SCI/NORM. In SD mode M+ enters DATA and SHIFT M+ deletes it; $\bar{x}$, σn, $\sigma n{-}1$, Σx^2, Σx, n are SHIFT 7/8/9/4/5/6. a b/c fractions with d/c conversion, constant calculations (K) on repeated operators.

Planned (not yet working)

- Linear regression mode
- Base-N mode
- Replay
- Equation solver
- Two-line display
- Answer history export
- Speech read-out of results
- Colour-blind key theme
- Exam lock
- Custom shortcut keys

U-45 CALCULATORS models are independent software recreations; they are not affiliated with any calculator manufacturer and carry no third-party trademarks.

Appendix E — Quick-reference card



#1 SHIFT	**#2 MODE**	**#3 x^2 ⇧ $\sqrt{}$**
#4 log ⇧ 10x	**#5 ln** ⇧ ex	**#6 ON**
#7 ab/c ⇧ d/c	**#8 °′″** ⇧ ←	**#9 hyp**
#10 sin ⇧ $\sin^{-1}$	**#11 cos** ⇧ $\cos^{-1}$	**#12 tan** ⇧ $\tan^{-1}$
#13 +/− ⇧ $\sqrt[3]{}$	**#14 ►** ⇧ x^3	**#15 [(** ⇧ X↔Y
#16)] ⇧ 1/x	**#17 xy** ⇧ x1/y	**#18 MR** ⇧ Min
#19 7 ⇧ $\bar{x}$	**#20 8** ⇧ σn	**#21 9** ⇧ σn−1
#22 C ⇧ X↔M	**#23 AC** ⇧ SAC	**#24 4** ⇧ Σx^2
#25 5 ⇧ Σx	**#26 6** ⇧ n	**#27 ×** ⇧ ENG←
#28 ÷ ⇧ ENG	**#29 1** ⇧ nPr	**#30 2** ⇧ nCr
#31 3 ⇧ x!	**#32 +** ⇧ R→P	**#33 −** ⇧ P→R
#34 0 ⇧ RND	**#35 •** ⇧ RAN#	**#36 EXP** ⇧ π
#37 = ⇧ %	**#38 M+** ⇧ M−	

U-45 CALCULATORS · Calculator #23 · cut out and keep